

A Corrected Conditional Reduction for the Non-Monotone Mean-Based No-Regret Conjecture

Prepared after audit of the submitted draft

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Abstract

Status. This note does not give an unconditional solution of Conjecture 3.12 of Muthukumar–Phade–Sahai. The submitted draft contained serious mathematical errors, the most important being an invalid upgrade from “not almost-sure convergence” to probability-zero convergence and an off-by-one shift inside a non-monotone strategy map. After correcting those points, the valid conclusion is conditional: the conjecture follows from a central local probability-mass lower bound, i.e. from the same “shaky-hands” obstruction isolated in the source paper.

The main contribution of this write-up is a clean re-indexed proof of the conditional implication, with no monotonicity and no hidden shift from $\hat{Q}_n$ to $\hat{Q}_{n-1}$. A separate theorem states exactly what extra uniform prefix estimate would be needed to obtain the stronger probability-zero conclusion claimed in the submitted draft. That extra estimate is not proved here.

1 Outcome of the audit

Let

$$E := \{ \lim_{t \rightarrow \infty} (P_t, Q_t) \text{ exists} \}.$$

The actual conjecture in Muthukumar–Phade–Sahai is that, for mean-based not necessarily monotone uniformly no-regret strategies with regret rate $(1/2, c)$, the pair (P_t, Q_t) does *not* converge almost surely. This means that one must rule out $\mathbb{P}(E) = 1$. The stronger statement $\mathbb{P}(E) = 0$ is not equivalent to the conjecture.

The submitted draft tried to prove $\mathbb{P}(E) = 0$ by conditioning on a high-density finite cylinder h with $\mathbb{P}(E \mid h)$ close to one and then applying a proposition whose hypothesis was almost-sure convergence under $\mathbb{P}(\cdot \mid h)$. This is invalid: $\mathbb{P}(E \mid h) > 1 - \eta$ does not imply $\mathbb{P}(E \mid h) = 1$. Moreover, the constant η produced by the proposition depended on h , so the density-point argument did not justify choosing a cylinder satisfying the needed inequality. A second problem was the step from Z_n/n to $Z_{n-1}/(n-1)$ inside a non-monotone response map. Without monotonicity or regularity, this shift is not harmless.

The repair below avoids both mistakes. We re-index the response maps so that the count used in the proof is exactly the count seen by the next mixed action. We then prove the following conditional implication: if the dependent empirical count distribution has a central local lower bound comparable to an equilibrium binomial law, then almost-sure convergence is impossible. This is partial progress only; the local lower bound itself remains the hard unproved ingredient.

2 Setup and indexing convention

We consider a competitive 2×2 repeated game with action set $\{0, 1\}$ for both players and unique completely mixed Nash equilibrium $(p^*, q^*) \in (0, 1)^2$. The realized actions are $I_t, J_t \in \{0, 1\}$ and the mixed actions are $P_t, Q_t \in [0, 1]$.

The mean-based assumption says that player 1's mixed action after observing n actions of player 2 is a deterministic function of the empirical count of those n actions. To avoid the off-by-one ambiguity in the submitted draft, define response maps

$$F_n(x) := f_{n+1}(x), \quad G_n(x) := g_{n+1}(x), \quad n \geq 0,$$

so that

$$P_{n+1} = F_n\left(\frac{Z_n^Q}{n}\right), \quad Q_{n+1} = G_n\left(\frac{Z_n^P}{n}\right),$$

where

$$Z_n^Q := \sum_{s=1}^n J_s, \quad Z_n^P := \sum_{s=1}^n I_s.$$

The value at $n = 0$ is irrelevant; all asymptotic statements are for $n \rightarrow \infty$.

After relabelling actions if necessary, player 1's payoff advantage of action 1 over action 0 has the sign of $q - q^*$. This is the orientation in which the fluctuation-sensitivity statement below has an upper-deviation form. The opposite orientation is handled by interchanging the displayed upper and lower inequalities.

3 Deterministic ingredients from optimal no-regret

We use two consequences of the source paper. They are stated here in the indexing convention above.

Lemma 3.1 (time-average convergence). *For two uniformly no-regret players with regret rate $(1/2, c)$ in a competitive 2×2 game, the averages of mixed play satisfy*

$$\left| \frac{1}{T} \sum_{t=1}^T P_t - p^* \right| + \left| \frac{1}{T} \sum_{t=1}^T Q_t - q^* \right| \leq \frac{C}{\sqrt{T}}$$

for a constant $C < \infty$ depending only on the game and regret constants. Consequently, on every sample path on which (P_t, Q_t) converges, its limit is (p^*, q^*) .

Proof. This is Lemma 3.7 of Muthukumar–Phade–Sahai, rewritten for the present notation. The final assertion follows because Cesaro averages of a convergent sequence have the same limit. $\square$

Lemma 3.2 (fluctuation sensitivity). *Fix $0 < \delta < \min\{p^*, 1 - p^*\}/3$. There are constants $\alpha < \infty$ and infinitely many pairs of integers (n_k, s_k) with*

$$1 \leq s_k \leq \alpha \sqrt{n_k}$$

such that, for $B_m^{(q^*)} \sim \text{Bin}(m, q^*)$,

$$\mathbb{E} F_{n_k} \left(\frac{B_{n_k - s_k}^{(q^*)} + s_k}{n_k} \right) \geq p^* + 2\delta. \quad (1)$$

Likewise, possibly after passing to another infinite sequence,

$$\mathbb{E} F_{n_k} \left(\frac{B_{n_k-s_k}^{(q^*)}}{n_k} \right) \leq p^* - 2\delta. \quad (2)$$

Proof. This is Proposition 3.1 of Muthukumar–Phade–Sahai, with $F_n = f_{n+1}$ so that the response map is evaluated on exactly the empirical average observed before the next move. A finite index shift changes the regret bound only by an additive constant and does not affect the existence of the infinite sequence. $\square$

Define the upper bad set associated with (1) by

$$A_k^+ := \left\{ z \in \{0, \dots, n_k\} : F_{n_k} \left(\frac{z}{n_k} \right) \geq p^* + \delta \right\}.$$

Define the lower bad set A_k^- analogously by replacing $\geq p^* + \delta$ with $\leq p^* - \delta$.

Lemma 3.3 (mass of the bad sets). *Let ν_k^+ be the law of $B_{n_k-s_k}^{(q^*)} + s_k$. If (1) holds, then*

$$\nu_k^+(A_k^+) \geq \eta_+, \quad \eta_+ := \frac{\delta}{1 - p^* - \delta} > 0.$$

Let ν_k^- be the law of $B_{n_k-s_k}^{(q^)}$. If (2) holds, then*

$$\nu_k^-(A_k^-) \geq \eta_-, \quad \eta_- := \frac{\delta}{p^* - \delta} > 0.$$

Proof. For the upper case,

$$\begin{aligned} p^* + 2\delta &\leq \sum_z F_{n_k}(z/n_k) \nu_k^+(z) \\ &\leq (p^* + \delta)(1 - \nu_k^+(A_k^+)) + \nu_k^+(A_k^+), \end{aligned}$$

which rearranges to the claimed lower bound. For the lower case, put $A = A_k^-$ and $Y \sim \nu_k^-$. Since $F_{n_k} \geq 0$ on A and $F_{n_k} > p^* - \delta$ on A^c ,

$$p^* - 2\delta \geq \mathbb{E} F_{n_k}(Y/n_k) \geq (p^* - \delta) \mathbb{P}\{Y \notin A\},$$

and the claimed bound follows. $\square$

We also need a simple comparison between an equilibrium binomial law and the shifted laws in Lemma 3.2.

Lemma 3.4 (shifted-binomial comparison). *Fix $q \in (0, 1)$, $\alpha < \infty$, and $\Gamma < \infty$. There are constants $c_+(q, \alpha, \Gamma) > 0$ and $c_-(q, \alpha, \Gamma) > 0$ such that, for all sufficiently large n , all integers s with $0 \leq s \leq \alpha\sqrt{n}$, and all integers z with $|z - nq| \leq \Gamma\sqrt{n}$,*

$$\mathbb{P}\{\text{Bin}(n, q) = z\} \geq c_+ \mathbb{P}\{\text{Bin}(n - s, q) + s = z\}, \quad (3)$$

and

$$\mathbb{P}\{\text{Bin}(n, q) = z\} \geq c_- \mathbb{P}\{\text{Bin}(n - s, q) = z\}. \quad (4)$$

Proof. For the shifted-up law, whenever the denominator is nonzero,

$$\frac{\mathbb{P}\{\text{Bin}(n, q) = z\}}{\mathbb{P}\{\text{Bin}(n - s, q) + s = z\}} = q^s \prod_{\ell=0}^{s-1} \frac{n - \ell}{z - \ell}.$$

For $|z - nq| \leq \Gamma\sqrt{n}$ and $s \leq \alpha\sqrt{n}$, the last display is at least

$$\left(\frac{q(n - s)}{nq + \Gamma\sqrt{n}} \right)^s \geq \left(\frac{1 - \alpha/\sqrt{n}}{1 + \Gamma/(q\sqrt{n})} \right)^{\alpha\sqrt{n}},$$

which is bounded below by a positive constant for all large n . This proves (3). Similarly,

$$\frac{\mathbb{P}\{\text{Bin}(n, q) = z\}}{\mathbb{P}\{\text{Bin}(n - s, q) = z\}} = (1 - q)^s \prod_{\ell=0}^{s-1} \frac{n - \ell}{n - z - \ell}$$

and, using $n - z \leq n(1 - q) + \Gamma\sqrt{n}$, this ratio is bounded below by a positive constant for all large n . This proves (4). $\square$

4 The corrected conditional implication

The following is the local probability-mass lower bound needed in the non-monotone argument. It is a conditional, central-window version of the source paper's shaky-hands property.

Definition 4.1 (central local minorisation after a prefix). Fix a finite history cylinder h with $\mathbb{P}(h) > 0$, and let $\mathbb{P}_h$ denote probability conditional on h . We say that player 2's empirical count satisfies central local minorisation after h if, for every $\Gamma < \infty$, there are constants $\rho_h(\Gamma) > 0$ and $N_h(\Gamma) < \infty$ such that for all $n \geq N_h(\Gamma)$ and all sets

$$A \subseteq \{z \in \{0, \dots, n\} : |z - nq^*| \leq \Gamma\sqrt{n}\},$$

one has

$$\mathbb{P}_h\{Z_n^Q \in A\} \geq \rho_h(\Gamma) \mathbb{P}\{\text{Bin}(n, q^*) \in A\}. \quad (5)$$

Remark 4.2. A pointwise lower bound

$$\mathbb{P}_h\{Z_n^Q = z\} \geq \rho_h \mathbb{P}\{\text{Bin}(n, q^*) = z\}$$

for all central z implies Definition 4.1. The source paper's Definition 3.13 compares Z_n^Q to a sum of independent Bernoulli variables with parameters $\mathbb{E}Q_s$; its Appendix C then compares that independent sum to $\text{Bin}(n, q^*)$ under the convergence-in-probability hypotheses. Definition 4.1 packages exactly the local lower bound needed for the non-monotone response-map argument.

Proposition 4.3 (conditional oscillation from local minorisation). *Let h be a finite history cylinder with $\mathbb{P}(h) > 0$. Suppose that, under $\mathbb{P}_h$,*

$$P_t \rightarrow p^*, \quad Q_t \rightarrow q^* \quad \text{almost surely}, \quad (6)$$

and suppose that the central local minorisation condition of Definition 4.1 holds after h . Then (6) is impossible.

Proof. We prove the upper-deviation case; the lower-deviation case is identical, using A_k^- . Choose $0 < \delta < \min\{p^*, 1 - p^*\}/3$ and take the sequence (n_k, s_k) from Lemma 3.2. Discard finitely many k so that n_k is larger than the length of h . Lemma 3.3 gives $\nu_k^+(A_k^+) \geq \eta_+ > 0$.

Choose Γ so large that

$$\nu_k^+\{|z - n_k q^*| > \Gamma\sqrt{n_k}\} \leq \eta_+/2$$

for all sufficiently large k . This is possible by Chebyshev's inequality, since $s_k = O(\sqrt{n_k})$. Hence, with

$$W_k := \{z : |z - n_k q^*| \leq \Gamma\sqrt{n_k}\},$$

we have

$$\nu_k^+(A_k^+ \cap W_k) \geq \eta_+/2 \quad (7)$$

for all large k .

By Lemma 3.4, there is $c_+ > 0$ such that

$$\mathbb{P}\{\text{Bin}(n_k, q^*) \in A_k^+ \cap W_k\} \geq c_+ \nu_k^+(A_k^+ \cap W_k) \geq c_+ \eta_+/2 \quad (8)$$

for all large k . Applying central local minorisation after h to $A = A_k^+ \cap W_k$ yields

$$\mathbb{P}_h\{Z_{n_k}^Q \in A_k^+\} \geq \rho_h(\Gamma) c_+ \eta_+/2 \quad (9)$$

for all large k .

On the event $\{Z_{n_k}^Q \in A_k^+\}$, the indexing convention gives exactly

$$P_{n_k+1} = F_{n_k}(Z_{n_k}^Q/n_k) \geq p^* + \delta.$$

Therefore

$$\mathbb{P}_h\{P_{n_k+1} \geq p^* + \delta\} \geq \rho_h(\Gamma) c_+ \eta_+/2$$

for all large k . This contradicts $P_t \rightarrow p^*$ almost surely under $\mathbb{P}_h$, because almost-sure convergence would imply

$$\mathbb{P}_h\{P_{n_k+1} \geq p^* + \delta\} \rightarrow 0$$

by dominated convergence. $\square$

Theorem 4.4 (corrected conditional form of the conjecture). *Assume the following conditional shaky-hands principle: whenever a finite history h has positive probability and the process conditioned on h satisfies $P_t \rightarrow p^*$ and $Q_t \rightarrow q^*$ almost surely, the central local minorisation condition of Definition 4.1 holds after h . Then Conjecture 3.12 holds: the pair (P_t, Q_t) cannot converge almost surely.*

Proof. Suppose for contradiction that (P_t, Q_t) converges almost surely. By Lemma 3.1, its limit is (p^*, q^*) almost surely. Apply Proposition 4.3 with the empty prefix $h = \emptyset$. This gives a contradiction. $\square$

Remark 4.5 (what has, and has not, been proved). Theorem 4.4 is a repaired conditional proof. It does not prove the central local minorisation condition. Thus it is not an unconditional solution of the authors' problem. It is best viewed as a rigorous restatement of the source paper's shaky-hands reduction, with the indexing and measure-theoretic issues in the submitted draft removed.

5 Why the probability-zero upgrade needs more

The submitted draft attempted to show $\mathbb{P}(E) = 0$. The conditional result above proves only that $\mathbb{P}(E) \neq 1$ under the local minorisation input. This is exactly the level required by Conjecture 3.12. The stronger probability-zero statement requires a uniform anti-density property over all finite prefixes.

Proposition 5.1 (non-convergence in probability does not imply probability-zero convergence). *There exists a stochastic process (X_t) such that X_t fails to converge in probability to 0, but the event $\{X_t \rightarrow 0\}$ has probability $1/2$.*

Proof. Let U be a Bernoulli($1/2$) random variable. Set $X_t = 0$ for all t on $\{U = 0\}$, and set $X_t = (-1)^t$ for all t on $\{U = 1\}$. Then $\mathbb{P}\{|X_t| \geq 1/2\} = 1/2$ for every t , so X_t does not converge to 0 in probability. But $X_t \rightarrow 0$ on the event $\{U = 0\}$, which has probability $1/2$. $\square$

Definition 5.2 (uniform cylinder oscillation). The process has uniform cylinder oscillation if there exist constants $\delta > 0$ and $\varepsilon > 0$ such that for every finite history cylinder h with $\mathbb{P}(h) > 0$,

$$\mathbb{P}_h \left(\limsup_{t \rightarrow \infty} |P_t - p^*| \geq \delta \right) \geq \varepsilon. \quad (10)$$

Theorem 5.3 (a valid probability-zero upgrade). *If Lemma 3.1 holds and the process has uniform cylinder oscillation, then*

$$\mathbb{P}\{\lim_{t \rightarrow \infty} (P_t, Q_t) \text{ exists}\} = 0.$$

Proof. Let $E = \{\lim_t (P_t, Q_t) \text{ exists}\}$. By Lemma 3.1, on E the limit is (p^*, q^*) . Therefore E is disjoint from

$$D_\delta := \left\{ \limsup_{t \rightarrow \infty} |P_t - p^*| \geq \delta \right\}.$$

Suppose $\mathbb{P}(E) > 0$. Let $(\mathcal{F}_n)$ be the finite-history filtration. By the martingale convergence theorem,

$$\mathbb{P}(E \mid \mathcal{F}_n) \rightarrow \mathbf{1}_E$$

almost surely. Hence there is a finite cylinder h with $\mathbb{P}(h) > 0$ and

$$\mathbb{P}_h(E) > 1 - \varepsilon/2.$$

Since $D_\delta \cap E = \emptyset$, this gives $\mathbb{P}_h(D_\delta) < \varepsilon/2$, contradicting (10). Thus $\mathbb{P}(E) = 0$. $\square$

Corollary 5.4 (one sufficient condition for probability-zero convergence). *Suppose the central local minorisation of Definition 4.1 holds with a lower-bound constant $\rho(\Gamma) > 0$ independent of the finite prefix h . Then the probability-zero conclusion of Theorem 5.3 follows.*

Proof. Repeat the proof of Proposition 4.3 after an arbitrary prefix h . Because the minorisation constant is independent of h , the lower bound in (9) gives the same positive constants δ and ε for every cylinder. This is uniform cylinder oscillation, so Theorem 5.3 applies. $\square$

Remark 5.5. This corollary explains exactly why the submitted proof of $\mathbb{P}(E) = 0$ failed. A prefix-dependent lower bound $\varepsilon_h > 0$ is not enough for the density-point argument; one needs a fixed positive lower bound that survives conditioning on arbitrary finite histories, or another argument of comparable strength.

6 Conclusion

After repair, the status is as follows.

1. The submitted manuscript is not a solution. Its claimed probability-zero theorem is not proved, and the non-monotone off-by-one shift is invalid.
2. A rigorous conditional theorem is available: central local minorisation of the empirical count distribution implies that almost-sure convergence is impossible. This is the correct level of conclusion for Conjecture 3.12.
3. The remaining obstacle is precisely the shaky-hands/local-PMF lower bound. Proving that bound for arbitrary non-monotone mean-based optimal no-regret strategies would complete the conjecture.
4. The stronger claim $\mathbb{P}(E) = 0$ requires a genuinely stronger, uniform prefix version of the local lower bound. That stronger statement is isolated in Theorem 5.3 and Corollary 5.4, but it is not established here.

Thus the result of this write-up is **partial progress, not a full solution**.

References

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